HDOC DAY – 62

Algorithm

1. Start.
2. Read the size and elements of arr1[].
3. Read the size and elements of arr2[].
4. Pass the addresses of both arrays and their sizes to merge_sorted_arrays().
5. Declare merged[] of size size1 + size2.
6. Initialize i = 0, j = 0, and k = 0.
7. Compare the elements at positions i and j of the two arrays.
8. Store the smaller element in merged[k] and increment the corresponding pointer.
9. Increment k.
10. Repeat until one of the arrays is completely traversed.
11. Copy the remaining elements of arr1[], if any, into merged[].
12. Copy the remaining elements of arr2[], if any, into merged[].
13. Display all elements of merged[].
14. Stop.

C PROGRAM:

```
#include <stdio.h>


void merge_sorted_arrays(int *arr1, int size1, int *arr2, int size2)
{
   int merged[size1 + size2];
   int i = 0, j = 0, k = 0;
```

```
while (i < size1 && j < size2)
{
   if (*(arr1 + i) <= *(arr2 + j))
   {
      merged[k] = *(arr1 + i);
      i++;
   }
   else
   {
      merged[k] = *(arr2 + j);
      j++;
   }
   k++;
}

while (i < size1)
{
   merged[k] = *(arr1 + i);
   i++;
   k++;
}

while (j < size2)
{
```

```
        merged[k] = *(arr2 + j);
        j++;
        k++;
    }

    for (i = 0; i < size1 + size2; i++)
        printf("%d ", merged[i]);

    printf("\n");
}

int main()
{
    int size1, size2, i;

    printf("Enter the size of first array: ");
    scanf("%d", &size1);

    int arr1[size1];

    printf("Enter the elements of first array: ");
    for (i = 0; i < size1; i++)
        scanf("%d", &arr1[i]);

    printf("Enter the size of second array: ");
```

```
    scanf("%d", &size2);

    int arr2[size2];

    printf("Enter the elements of second array: ");
    for (i = 0; i < size2; i++)
        scanf("%d", &arr2[i]);

    merge_sorted_arrays(arr1, size1, arr2, size2);

    return 0;
}
```

TEST CASE TABLE:

Test Case	Array 1	Array 2	Expected Output
1	1 3 5 7	2 4 6 8	1 2 3 4 5 6 7 8
2	1 2 2 4	2 3 5	1 2 2 2 3 4 5
3	1 4 7	2 3 5 6 8	1 2 3 4 5 6 7 8
4	1 2 3	4 5 6	1 2 3 4 5 6
5	2 2 5	1 2 4	1 2 2 2 4 5